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Laminar burning velocities of hydrogen-oxygen-steam mixtures at elevated temperatures and pressures

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Abstract

Effects of elevated temperatures (up to 573 K) and pressures (up to 72 bar) on the laminar flame velocity of hydrogen-oxygen-steam mixtures were studied both experimentally and computationally. Stoichiometric hydrogen-oxygen mixtures diluted with steam up to 85% (mol.) have been tested in a spherical explosion chamber with an inner diameter of 25 cm using the pressure method and high speed shadow cinematography. The experimental data on the laminar burning velocity were compared with numerical calculations that used different H/O reaction mechanisms based on the Lutz, GRI-Mech 3.0, Li and Warnatz schemes. The calculations were performed with three codes: INSFLA, Cantera (for unstretched planar flames) and our own FP-code (based on the PREMIX code). Nonmonotonic pressure dependence and strong suppressing effect of steam dilution on laminar flame velocity was found both experimentally and numerically. The best consistency with the experimental measurements of the laminar flame velocity at elevated pressures and temperatures in presence of high steam concentrations was found for the Lutz H/O mechanism.
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Keywords: Laminar burning velocity; Hydrogen-oxygen-steam mixtures; Radiolysis gas

1. Introduction

The laminar burning velocity (LBV) is an important overall characteristic of the reactivity of combustible mixtures. It is usually used as a measure of the combustion rate in numerical models to predict the dynamics of combustion. Laminar burning velocity measurements can be used to evaluate the laminar flame thickness δ, the overall activation energy E_a, and the reaction order n for the dimensionless analysis of a turbulent combustion and to estimate the flame stability and flame acceleration potential.

According to the Zeldovich–Frank–Kamenetskii theory [1], the overall activation energy E_a represents the sensitivity of the laminar burning rate at a fixed pressure to the flame temperature as follows

$$S_L(T_b) \propto S_L \cdot \exp\left(-\frac{E_a}{2 \cdot T_b}\right), \quad (1)$$

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where $S_L(T_b)$ and S_L are laminar burning velocities at a given flame temperature T_b and at normal initial conditions (STP). The pressure dependence of the laminar velocity as a function of the reaction order n at fixed flame temperature can be written in a simplified form:

$$S_L(p) = S_L(p_0) \cdot \left(\frac{p}{p_0}\right)^{\frac{n}{2}-1} \quad (2)$$

where $S_L(p)$ and $S_L(p_0)$are laminar burning velocities at a given pressure p compared to the standard pressure $p_0 = 1$ bar. The overall reaction order n indicates the pressure influence on the combustion rate. For instance, if the reaction order $n < 2$, the laminar burning velocity will decrease with an initial pressure increase.

Hydrogen-oxygen-steam combustion at different pressures and temperatures is an important issue from a practical point of view. For instance, radiolysis gas consisting of a stoichiometric hydrogen-oxygen mixture diluted with steam at temperatures of up to 573 K and pressures of up to 70 bar is a typical gas composition relevant for the piping system of Boiling Water Reactors (BWR) in Nuclear Power Plants (NPP). Such a system also allows to study the effect of a diluent like steam on the combustion rate of the highly reactive stoichiometric hydrogen-oxygen mixture using the simplest H/O oxidation mechanism. One of the problems to be expected is that the experimental conditions of the current work are quite close to the self-ignition limits of the H_2–O_2–H_2O mixture. Other problems to be technically solved are that the maximum combustion pressure for some mixtures reaches 800 bar with a maximum visible flame velocity of 100–120 m/s which can generate relatively strong shock waves of Mach number $M = 1.2$ with over-pressure of $\Delta p/p_0 = 0.3$.

Experimental data on the laminar burning velocity of hydrogen-oxygen mixtures with ratios of hydrogen to oxygen ranging from 0.2 to 9 were obtained by Koroll et al. at a temperature of 298 K and a pressure of 101.3 kPa [2]. The burning velocity of a stoichiometric hydrogen-oxygen mixture was measured to be 10.7 m/s at such conditions. Laminar burning velocities for stoichiometric hydrogen-oxygen mixtures diluted with steam were experimentally evaluated in Koroll's work [3] using a burner method at normal pressure and at an elevated temperature of 373 K.

Some authors, like Burke et al. [4] and Tse et al. [5], have also investigated the laminar flame speed of hydrogen-oxygen mixtures diluted with inert gases as nitrogen, argon and helium at high pressures. The authors found that the nonmonotonic pressure dependence of the laminar flame speed might be connected with a change of the reaction mechanism due to the more important role of three-body reactions at higher pressures, thereafter the termolecular $H + O_2$ (+M) chain termination reaction starts to dominate. This leads to the lower overall reaction order with decreasing laminar flame velocity at higher pressures. Main difference with the current work will be that due to the steam dilution the water molecule takes part not only as an inert agent in three-body reactions but also as a product in chain-branching reaction $H_2 + OH = H_2O + H$ which can be suppressed in the case of high steam dilutions. The same nonmonotonic behavior against an initial pressure was also found for the flammability [6,7] and self-ignition limits [7] of hydrogen-oxygen-steam mixtures. A specific point at pressure of 7 bar with lower steam dilution in Fig. 1 might be associated with a change of chain-branching kinetics to straight-chain kinetics.

There is no experimental data on the laminar flame velocity for stoichiometric hydrogen-oxygen mixtures diluted with steam (or so called radiolysis gas mixtures) at elevated pressures of up to 72 bar and temperatures of up to 573 K typical for Boiling Water Reactors. The objective of the current work was to obtain experimental data on the laminar burning velocities in stoichiometric hydrogen-oxygen mixtures diluted with steam at the BWR relevant initial conditions. With these data the possibility of combustion events and their regimes can be evaluated for Nuclear Power Plant safety applications using computer codes capable to make detailed and reliable predictions.

2. Experimental details

The experiments on the laminar burning velocity were conducted in a spherical stainless steel explosion bomb with an internal diameter of 25 cm ($V = 8.2$ dm^3) a wall thickness more than 34 mm and two quartz windows for optical observations. The bomb was designed to withstand static pressure of up to 800 bar and temperature up to 300 °C (Fig. 2).

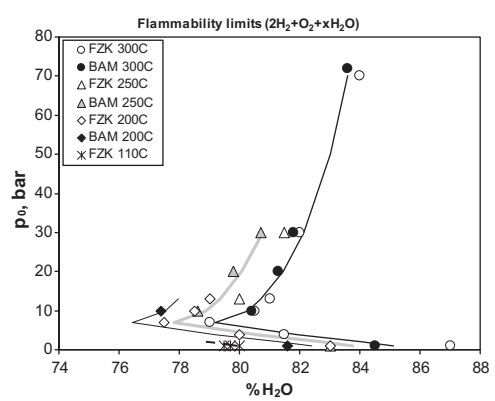

Fig. 1. Experimental flammability limits for stoichiometric hydrogen-oxygen–steam mixtures at elevated pressures and temperatures: BAM data [6]; FZK data [7].

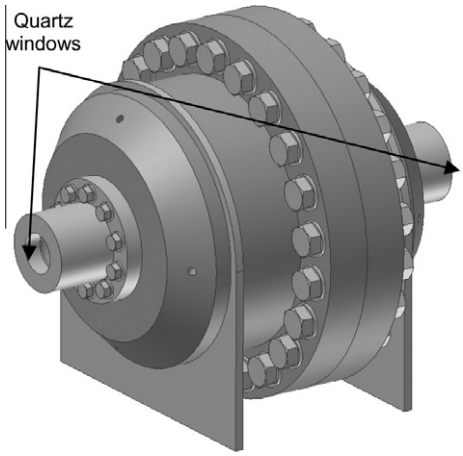

Fig. 2. Explosion bomb with quartz windows (shown without heating elements).

Stoichiometric hydrogen-oxygen mixtures diluted with steam were used in the tests. Taking into account a nonideality of the test mixtures with steam at pressures of about 70 bar, the gas injection system was based on the mass flow rate control. With mass flow control, the mole-concentration of the gas mixture is known without calculations from partial pressure and consideration of the compressibility factor $Z = p \cdot V/(R \cdot T) = 0.78$ for H_2O at a pressure of 70 bar according to the real gas equation of state based on the National Institute of Standards and Technology (NIST) tables [8]. Each component of a gas mixture was heated separately and then mixed just before the mixture was injected into the test volume. The mixture quality was also controlled using a gas analyzer via bypass line before the test volume was filled. The accuracy was better than ±0.5%.

Combustion experiments were performed at initial temperatures of 383, 473 and 573 K and pressures of 1, 4, 10, 13, 30 and 70(72) bar. In the experiments with steam, steam condensation processes had to be taken into account. Table 1 shows the maximum steam concentration which could be achieved for a given pressure and temperature. "100%" indicates that test mixtures with steam concentrations up to 100% can be prepared. Other values show that, for instance, at a temperature of 473 K and a pressure of 30 bar no mixtures with steam concentrations of more than 51.8% of steam can be prepared.

The gas temperature was measured using a NiCr–Ni thermocouple of 0.5 mm thickness that was installed in a distance of 6 cm from the wall inside the test mixture. The temperature control system kept the initial temperature with an accuracy of ±0.5 °C. The spatial non-uniformity of the temperature was less than ±1 °C. The steam concentration was changed in the range of 0–85%(mol.). A calibrated spark plug or glow plug was used to ignite the test mixture. For test mixtures with less than 60%(mol.) H_2O the ignition energy was less than 5 mJ. For some mixtures close to the flammability limit the minimum ignition energy could be about 0.5–1.0 J. At initial pressures higher than 10 bar a glow plug had to be used because the high voltage electric spark could not be properly isolated at high pressures in presence of steam atmosphere. The experiments showed that both igniters give reliable data on the flame velocity.

The measurements of laminar burning velocities were performed using high speed shadow cinematography (up to 10,000 fps) and pressure measurements. Two piezoelectric pressure transducers with an upper temperature limit of 350 °C and with a measuring range of 0–250 bar (Kistler 6001) and 0–5000 bar (Kistler 6012) were used to measure the combustion pressure.

3. Calculations

Numerical calculations of the laminar burning velocities were accomplished using a modified FP-code (based on the PREMIX program) [9], the INSFLA code [10] and the CANTERA code [11] for unstretched (planar) flames. Three different codes for the numerical calculations were used to achieve a convergence of the numerical solution in a proper time even for mixtures close to the flammability limits.

To calculate laminar burning velocities of hydrogen-oxygen-steam mixtures at the extremely high initial pressures and temperatures, different H/O reaction mechanisms based on the Lutz [12] and Li [13] schemes, the GRI-Mech 3.0 mechanism [14], and the Warnatz scheme, implemented into the INSFLA code [10], were used in this work. Li and Warnatz mechanisms have been

Table 1
Maximum steam concentration at experimental initial conditions.

Temperature (T, K)	Saturation pressure (ps, bar)	Maximum steam content (%H_2Omax)					
		1 bar	4 bar	10 bar	13 bar	30 bar	72 bar
293	0.023	2.34	0.58	0.23	0.18	0.08	0.03
383	1.43	100	35.8	14.3	11.0	4.8	1.99
473	15.55	100	100	100	100	51.8	21.6
573	85.92	100	100	100	100	100	100

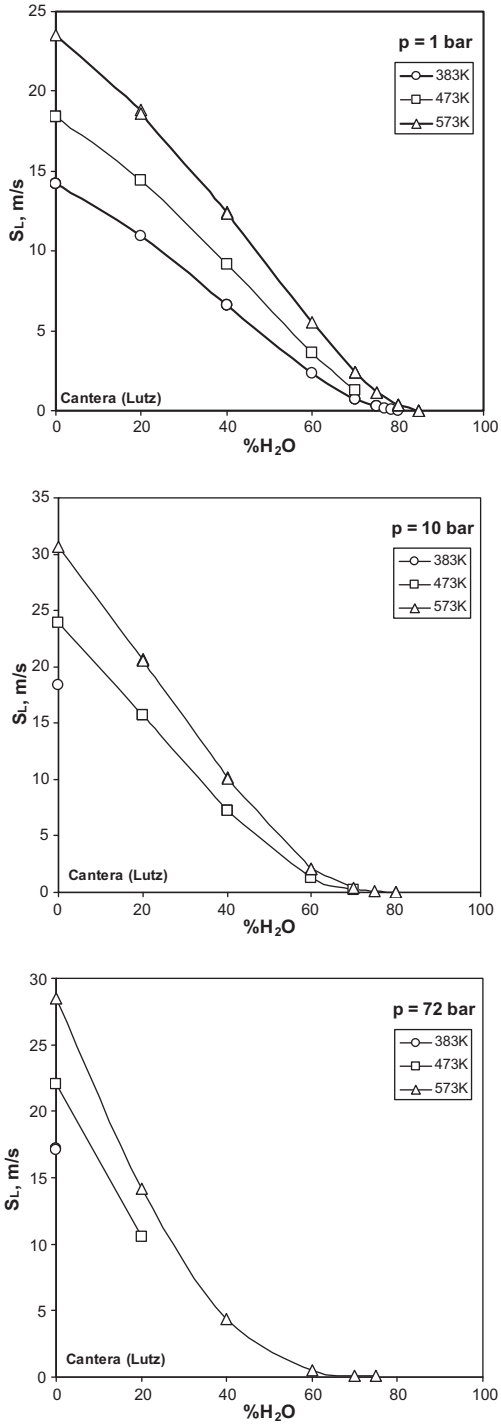

Fig. 3. Calculated laminar flame velocity vs. steam concentration for stoichiometric hydrogen-oxygen-steam mixtures (Lutz mechanism).

tested for high pressure combustion processes but not in presence of extremely high steam concentration like in current work.

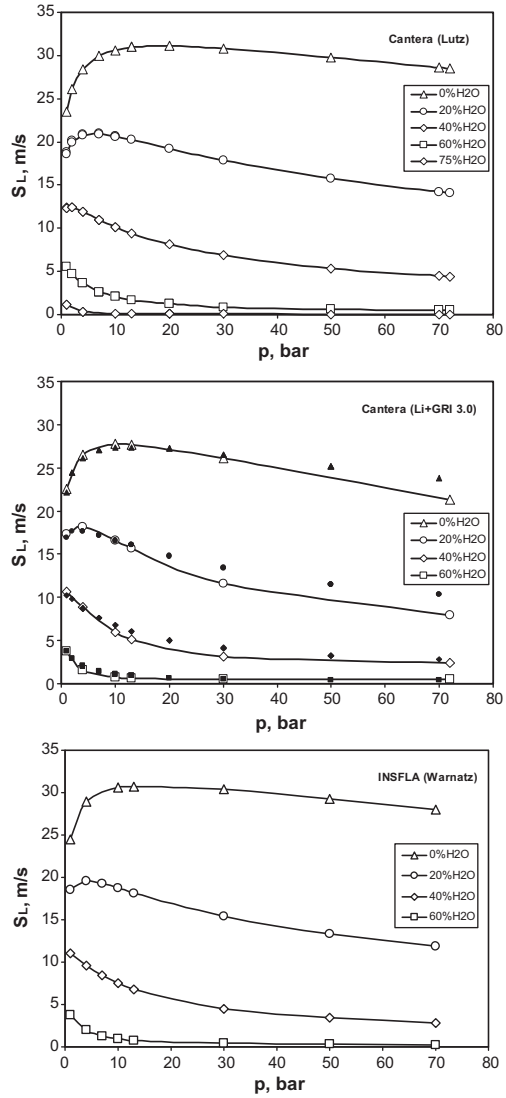

Fig. 4. Calculated laminar flame velocities vs. initial pressure for $H_2/O_2/H_2O$ mixtures ($T = 573$ K): (a) Lutz scheme; (b) Li (solid points) and GRI-Mech 3.0 (open points); (c) Maas–Warnatz mechanism.

Data of numerical calculations using these different mechanisms are summarized in Figs. 3 and 4 to illustrate the behavior of the laminar flame velocity of stoichiometric hydrogen-oxygen-steam mixtures at different initial conditions in the range of the variables used in current experiments.

The effect of steam dilution on the laminar burning velocity of hydrogen-oxygen-steam mixtures at different initial pressures is shown in Fig. 3. Steam dilution suppresses the combustion of hydrogen-oxygen mixtures. For instance, the laminar burning velocity reduces more than 100 times when the steam dilution is changed from 0

to 80 mol.% of steam. As it follows from Fig. 3, a higher initial temperature always has a promoting effect on the laminar burning velocity.

The nonmonotonic influence of the initial pressure on the laminar burning velocity of H_2–O_2–H_2O mixtures is shown in Fig. 4. This might be explained due to the change of chain-branching to straight-chain kinetics. The boundary between these two mechanisms associated with a maximum of the laminar flame velocity is shifted to lower pressures at the lower flame temperatures as stated in [4]. It was found that an increasing initial pressure has a promoting effect on the laminar burning velocity for mixtures with steam contents of not more than 20%. In this case, the flame velocity increases with a pressure increase from 1 to 10 bar. At an initial pressure of more than 10 bar the flame velocity decreases with pressure increase up to 72 bar. For hydrogen-oxygen mixtures diluted with more than 20% of steam, increasing initial pressure becomes a strong suppressing factor on the laminar burning velocity for all the pressures investigated. The laminar burning velocity becomes practically inversely proportional to the initial pressure. This means, the overall reaction order becomes much less than two for steam dilutions of 40 mol.% and more. For instance, for a hydrogen-oxygen mixture diluted with 75% H_2O the overall reaction order $n = 0.5$–0.8, which results in laminar flame velocity decrease from 100 to 5 cm/s for a pressure increase from 1 to 72 bar.

Calculations of laminar flame velocities were performed with all chosen mechanisms at different pressures, temperatures and steam dilutions. In general, all the mechanisms demonstrate the same behavior of the laminar flame velocity as a function of pressure. But compared to the Lutz H/O mechanism, the Li and GRI-Mech 3.0 mechanisms give lower absolute values of the flame velocity. For example for 60% of steam, the Lutz scheme gives a two times higher flame velocity at 573 K compared to the GRI-Mech 3.0 and Li mechanism. The Warnatz mechanism was found to be very sensitive to the steam concentration. It has an intermediate position between Lutz and GRI-Mech 3.0 mechanisms. At lower steam dilution it is close to the Lutz scheme, but at higher steam concentrations (>60 mol.%) the Warnatz mechanism gives lower values for the LBV than even the GRI-Mech-3.0 mechanism. The reason could be that the collision efficiency of third-body reactions for Warnatz mechanism is too high. Such different results mean, that the calculated data on the laminar flame velocity, obtained using different mechanisms, have to be experimentally validated to choose proper kinetics.

For the further safety analyses, the main combustion characteristics of hydrogen-oxygen-steam mixtures had to be calculated using numerical calculations of laminar burning velocities. To evaluate the overall reaction order n and the activation energy E_a, the variations of S_L with changed pressure or diluent concentration were used. In standard simplified form the overall activation energy can be calculated as follows

$$E_a = -2\frac{\partial \ln(\sigma S_L)}{\partial(1/T_b)}. \tag{3}$$

With known initial temperature T_u a dimensionless measure of the overall energy activation can also be calculated

$$Ze = \frac{T_b - T_u}{T_b^2}\frac{E_a}{R} \tag{4}$$

A dimensionless Markstein number as a measure of the flame stability and its sensitivity to the flame curvature is given by [15] as follows

$$Ma_b = (\sigma/(\sigma-1)) \cdot (\ln\sigma + (Ze(Le-1)/2/(\sigma-1)) \cdot \int_0^{\sigma-1} (\ln(1+x)/x)dx), \tag{5}$$

where Le is the Lewis number; x is the dummy variable of integration.

By equating E_a for two close ranges of p, the overall reaction order n is obtained. To evaluate the reaction order n for a small variation of the initial pressure p the following simplified equation was used at the same T_b:

$$n = 2\frac{\partial \ln(S_L)}{\partial \ln(p)} + 2. \tag{6}$$

The main combustion properties as a function of the steam dilution were calculated for hydrogen-oxygen-steam mixtures (see e.g. Table 2 for 1 bar and 573 K). As it follows from Table 2, the dry hydrogen-oxygen mixture (0% H_2O) at 573 K and different pressures has a visible flame velocity of more than 100 m/s. Such a flame velocity produces a leading shock wave and accelerates to a detonation very fast. It could be a big problem for the forthcoming LBV experiments because the tested stoichiometric hydrogen-oxygen mixture without steam could detonate even in the spherical explosion chamber. Two detonation events were recorded in the experiments. In the opposite case of lower mixture reactivity, the highly steam-diluted hydrogen-oxygen mixture (75% H_2O) has a visible flame velocity of about 2.8 m/s. Compared to the dry hydrogen-oxygen mixture, the steam-diluted mixture has much lower potential for the following flame acceleration and detonation transition.

The calculations show, that the overall reaction order decreases with steam dilution increase. This means, that the laminar burning velocity decreases with a pressure increase at higher steam dilution. The Markstein number initially slightly increases with steam dilution increase. Then, at

Table 2
Combustion characteristics of stoichiometric hydrogen-oxygen-steam mixtures at 573 K.

Diluent %H_2O	Laminar velocity S_L (m/s)	Expansion rate σ	Flame thickness δ (mm)	Adiabatic temperature T_p (K)	Reaction order n	Activation energy E_a (K)	Zeldovich number Ze	Markstein number Ma
p = 1 bar								
0	23.5	4.467	0.275	3110	2.70	11490	3.02	0.144
20	18.3	4.182	0.294	2899	2.56	8740	2.42	0.155
35	13.4	3.932	0.308	2696	2.41	7930	2.32	0.176
40	11.7	3.834	0.313	2614	2.35	7870	2.35	0.186
50	8.28	3.599	0.329	2417	2.18	8030	2.53	0.204
60	5.04	3.282	0.360	2153	1.91	9030	3.08	0.194
75	1.14	2.591	0.541	1619	0.80	15880	6.34	−0.764
p = 10 bar								
0	29.8	4.788	0.019	3440	2.44	13540	3.28	0.296
20	18.9	4.43	0.019	3148	2.14	14070	3.66	0.382
35	11.2	4.115	0.020	2873	1.82	15420	4.30	0.448
40	8.82	3.992	0.020	2764	1.67	16070	4.61	0.455
50	4.78	3.7	0.023	2507	1.24	17850	5.49	0.273
p = 72 bar								
0	24.9	5.063	0.002	3751	1.90	22800	5.15	0.572
20	11.4	4.63	0.002	3363	1.43	24950	6.16	0.779

higher steam dilution (>50% H_2O), it becomes negative. For the mixtures with more than 75% H_2O the Ma < 0 for all pressures investigated. The threshold for Ma < 0 shifts to lower steam concentrations at higher pressures. This means that the flame is going to be unstable at higher pressures with high steam dilution (>50% H_2O) and the flame development can sometimes lead to a local quenching which can prevent a subsequent flame acceleration and detonation transition [16].

4. Experimental results and discussion

The spherical bomb method developed by Andrews and Bradley [17] has been chosen as main tool for the experimental measurements of burning velocities in this work. The main and simplest technique to determine the burning velocity via this method is to measure the pressure rise against time at the initial stage of the spherical flame propagation. The problem was, that a fine pressure variation has to be measured very precisely in a relatively short time. For this purpose two pressure sensors with high and low pressure resolution were used. To evaluate the laminar burning velocity we used the dependence which takes into account the adiabatic change of pressure and temperature of the unburned material as follows [18]:

$$S_L = \frac{S_S}{\sigma}\left(1 + \frac{1}{\gamma_b}\frac{B_2 r_b^3}{(S_S^3 P_0 + B_2 r_b^3)}\right) \tag{7}$$

Here B_2 is the polynomial coefficient of the correlation of the experimental pressure–time history in the form

$$p(t) = p_0 + B_2 \cdot t^3, \tag{8}$$

where p_0 is the ambient pressure, r_b is the burned gas radius, $\sigma = \rho_u/\rho_b$ is the expansion rate of burned mixture, γ_b, γ_u are specific heat ratios for the burned and unburned gas and S_S is the visual flame velocity given by:

$$S_S = \left(\frac{B_2}{P_0}\left(\frac{\gamma_b R^3 + \frac{\gamma_u}{\sigma} r_b^3 - \gamma_b r_b^3}{(1-\frac{1}{\sigma})\gamma_b\gamma_u} - r_b^3\right)\right)^{1/3}, \tag{9}$$

where R is the internal radius of the explosion chamber.

An exemplary experimental curve of a pressure signal is shown in Fig. 5. The maximum combustion over-pressure in this particular case for 70 bar, 573 K, and 70% H_2O is estimated to be 121 bar. The initial part of the pressure–time record (Fig. 5, right) was used to calculate the experimental laminar flame velocity according to Eq. (7)–(9). Figure 5 (right) shows that the initial part of the experimental curve can be linearly approximated as a cubic root of the pressure dependence vs. time Eq. (8), where B_2 is the coefficient used to calculate the flame velocity according to the Eq. (7)–(9). These calculations give a visual flame velocity of $S_S = 25.4$ cm/sec and a normal laminar flame velocity of $S_L = 8.9$ cm/sec with an expansion ratio of $\sigma = 2.863$.

Only initial part of the experimental pressure record (less than $0.03 \cdot p_{max}$) was used for laminar flame velocity evaluation. With experimental data shown in Fig. 5 the laminar flame velocity can be calculated using an assumption about the isobaric combustion at the initial stage of combustion, when the flame velocity remains constant up to complete combustion of the test mixture. Using the same pressure–time correlation (8), the com-

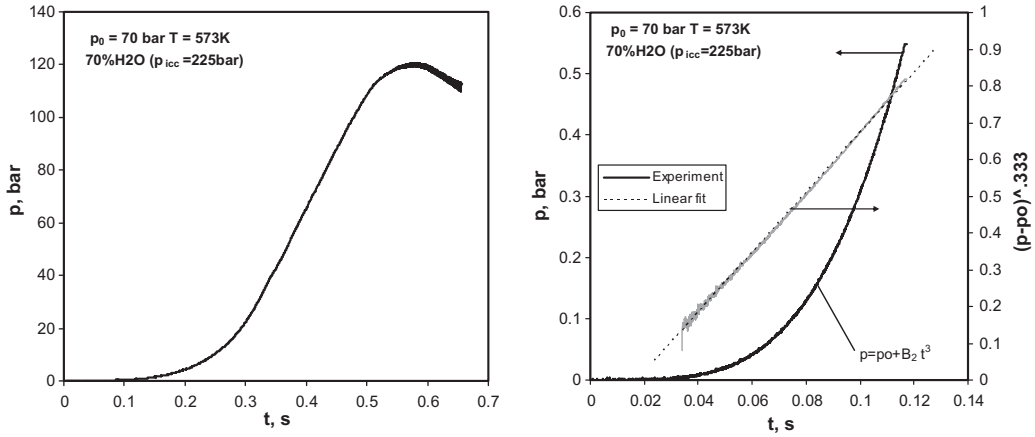

Fig. 5. Experimental pressure–time history of a combustion process (70 mol.% H_2O at 70 bar and 573 K) (left); zoomed view of initial part of the pressure–time history and its linear approximation (right).

plete burning time $t_{max} = ((p_{max} - p_0)/B_2)^{1/3}$, where p_{max} is the maximum pressure. For instance, in the case shown in Fig. 5 (left) $p_{max} = 120.7$ bar. The visual flame velocity in general is $S_S = R/t_{max}$, where $R = 12.5$ cm is the explosion chamber radius. Hence, in this case $S_S = 12.5/0.6 = 20.7$ cm/s. Taking into account the expansion ratio of the tested mixture of $\sigma = 2.863$, the laminar burning velocity $S_L = S_S/\sigma = 7.3$ cm/s. This value is consistent with the value $S_L = 8.9$ cm/s calculated using the Eq. (7)–(9) that were obtained from the classical laminar flame theory in a spherical case.

The high speed shadow technique was additionally used to evaluate the laminar flame velocity. Two examples for experimental sequences of high speed shadow photos are shown in Fig. 6. For relatively sensitive test mixtures, the flame has practically spherical shape. Since the Lewis number $Le = 1.06$ in both cases even at the pressure $p_0 = 30$ bar and steam concentration of 75 mol.% the flame initially looks stable without any cells. With an optical analysis, the maximum intensity gradient is between the dark and the bright zone on the recording plane. The maximum gradient position exactly identifies the outer border of the reaction zone. The experimental procedure consisted of the evaluation of the burnt zone radius in three directions (upward–downward, transverse, and 45°) versus time to define the laminar flame velocity.

The measurements of the laminar burning velocities were made using high speed shadow cinematography and pressure measurements. Figure 7

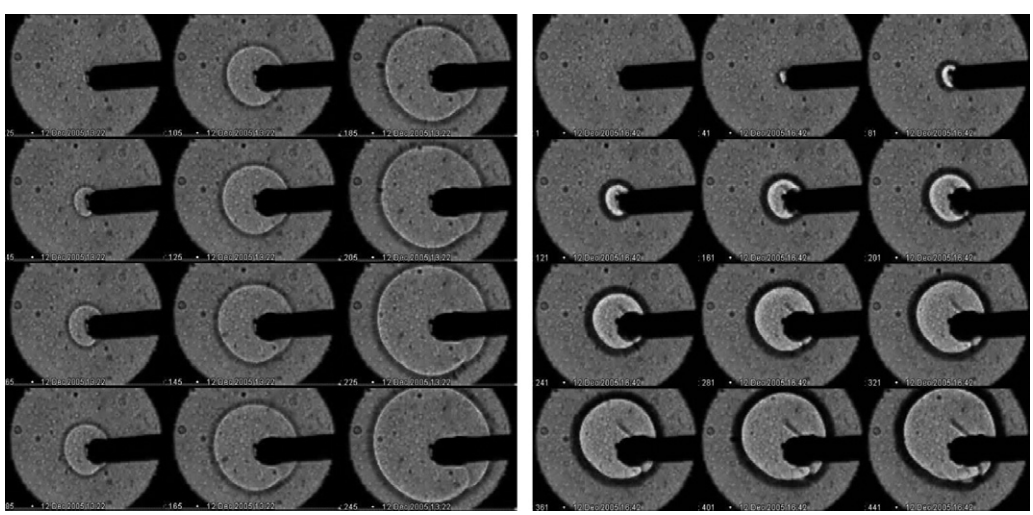

Fig. 6. High speed shadow photos of the combustion process: $p_0 = 4$ bar, $T = 573$ K, 75% H_2O, $\Delta t = 2$ ms (left); $p_0 = 30$ bar, $T = 573$ K, 75% H_2O $\Delta t = 16$ ms (right).

shows a comparison of laminar flame velocities measured using pressure and optical methods at an initial temperature of 573 K and different initial pressures for stoichiometric hydrogen-oxygen mixture with 75% of steam. As can be seen in Fig. 7, both experimental methods give practically the same results. This is very important for practical applications because it means that the flame shape can be assumed as an ideal sphere and the buoyancy effect can be neglected. Basically the pressure method was used for the experimental evaluation of the laminar flame velocity in this work.

It is very important to compare our experimental data with other measurements of laminar burning velocities. A comparison of the experimental data of the current work and Koroll's experimental data [3] at practically the same temperature is shown in Fig. 8. It demonstrates quite good agreement between the two data sets. In general, as it was shown in Fig. 8, all tested chemical kinetics mechanisms give good agreement with the experimental data for hydrogen-oxygen mixtures at rather low initial pressure (~1 bar) and initial temperature. Since the three experimental series in Fig. 8 repeated at the same initial pressure and demonstrated quite good agreement, the accuracy of the flame velocity evaluations was estimated to be ±5%. Compared to low pressure experiments for elevated temperatures and pressures (30 bar and 573 K in Fig. 9) the difference of the experimental data was found to be very significant for some of the mechanisms used. It was found that the Lutz mechanism gives surprisingly good agreement with experimental measurements independent of that the mechanism was developed for relatively low pressures of up to 2 bar. At low steam dilutions, the calculated LBV using Lutz and Li mechanisms, especially at higher pressure, are more consistent with the experimental data than the results obtained using GRI-Mech-3.0. A comparison of the experimental data with high

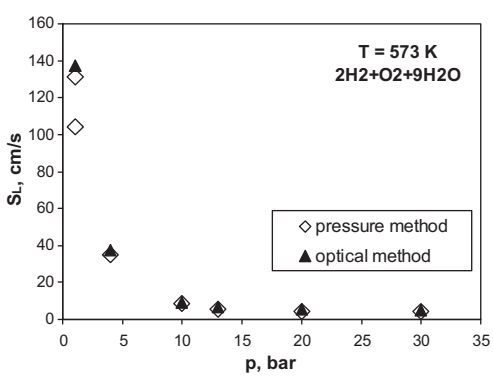

Fig. 7. Experimental data on laminar flame velocity obtained using pressure and optical methods for a 75% H_2O hydrogen-oxygen-steam mixture.

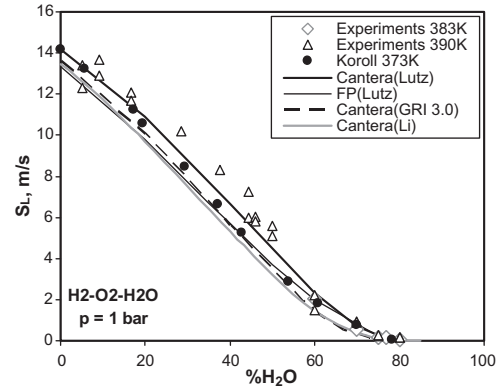

Fig. 8. Comparison of current experimental data on laminar burning velocity with Koroll's data [3] and numerical simulations (383 K, 1 bar).

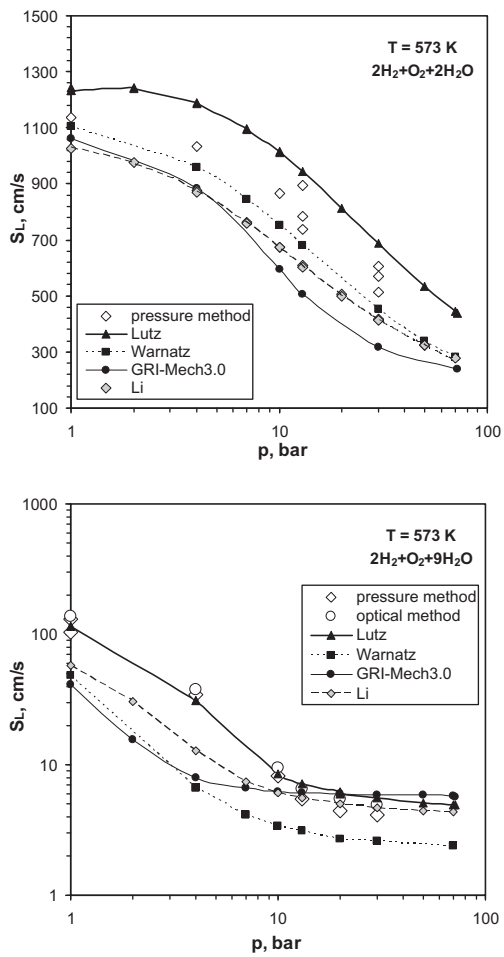

Fig. 9. Calculated and experimental laminar burning velocities for 40% (upper) and 75% H_2O (lower) stoichiometric hydrogen-oxygen-steam mixtures (573 K).

steam dilution of 40% and 75% H_2O at 573 K with the calculated data on the laminar flame velocities obtained using different chemical kinetics mechanisms shows better agreement for the Lutz and Maas–Warnatz schemes (Fig. 9). Only for 75% H_2O the Warnatz scheme gives two times lower flame velocities compared to the experimental data. The Li and GRI-Mech-3.0 mechanisms fit well only for higher initial pressures. At lower initial pressures in presence of high amount of steam, the velocities calculated with the Li and GRI-Mech-3.0 mechanisms can be two times lower than the experimental data.

Taking into account our data analysis we would recommend to use the Lutz H/O mechanism for further calculations of laminar burning velocities of hydrogen-oxygen-steam mixtures at elevated pressures and temperatures.

5. Concluding remarks

Two methods to obtain flame speeds were presented: the pressure method and an optical method based on a high speed shadow video technique. Flame speeds were determined for temperatures of 110, 200 and 300 °C at different pressures (up to 70 bar) and steam concentrations (0–80% H_2O). Laminar flame velocities varied from 25–30 m/s for pure hydrogen-oxygen to 2–4 cm/s for mixtures close to the flammability limit. High pressure and steam dilution have a suppressing effect on the flame propagation, while higher temperatures always promote the faster flame propagation.

Numerical calculations of laminar flame velocities using four different H/O chemical kinetics mechanisms were performed. Experimental validation of the numerical codes showed, that better agreement was achieved using the Lutz H/O mechanism. Overall kinetic characteristics were calculated for hydrogen-oxygen-steam mixtures against pressure, temperature and steam dilution. The flame instability threshold with negative Markstein number lies at lower steam dilutions with higher initial pressures.

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